

STATS 200A Lecture 4

Henry Liev

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Roll a fair die: $\Omega = \{1, \dots, 6\}, \mathcal{F} = 2^\Omega, \mathbb{P}(A) = |A|/6$

$$X(\omega) = \mathbf{1}\{\omega \text{ is odd}\}.$$

$$\{X \in (1/2, 2]\} = X^{-1}((-1/2, 2]) = \{1, 3, 5\}$$

$$X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$\{X \in B\} \quad B \in \mathcal{B}(\mathbb{R})$$

$$\mathbb{P}(\{X \in B\}) = \mathbb{P}(X^{-1}(B))$$

$$X^{-1}((-1/2, 2]) = \Omega$$

$$X^{-1}(B^c) = (X^{-1}(B))^c$$

$$\begin{array}{ccc} (\Omega, \sigma(\mathbf{X})) & \xrightarrow{X} & (E, \mathcal{E}) \\ \sigma(X) = X^{-1}(\mathcal{E}) & & \\ \text{(Initial / Pullback: \textbf{coarsest})} & & \end{array} \quad \begin{array}{ccc} (\Omega, \mathcal{F}) & \xrightarrow{X} & (E, \mathbf{X}_*\mathcal{F}) \\ X_*\mathcal{F} = \{B : X^{-1}(B) \in \mathcal{F}\} & & \\ \text{(Final: \textbf{finest})} & & \end{array}$$

$$X_i : (\Omega, ?) \rightarrow (E_i, \mathcal{E}_i)$$

$$(\Omega, 2^\Omega) \xrightarrow{X} (\mathbb{R}, \mathcal{B}(\mathbb{R})) \quad X^{-1}(\{1\}) = \{1, 3, 5\}$$

Definition 4.1 (initial σ -algebra for a family)

For maps $X_i : \Omega \rightarrow (E_i, \mathcal{E}_i)$,

$$\sigma(X_i : i \in I) = \sigma(\{X_i^{-1}(B) : i \in I, B \in \mathcal{E}_i\})$$

is the smallest σ -algebra making every X_i measurable

Definition 4.2

For measurable $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (E, \mathcal{E})$, the *law* of X is

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B) = \mathbb{P}(\{X \in B\}),$$

for $B \in \mathcal{E}$. Also $X_\# \mathbb{P}, \mathbb{P} \circ X^{-1}$: the *pushforward* $\mathbb{P}$ by X .

Proposition 4.3

$\mathbb{P}_X$ is a probability measure on $(E, \mathcal{E})$.